MODULE 5 URINARY SYSTEM

The Urinary System.

THE URINARY SYSTEM

Covers the gross anatomy of the kidney, the nephron, the juxtaglomerular apparatus and filtration membrane, the blood supply, and the ureters, bladder, and urethra. The focus is on the structures and the job each one does.

BY THE END

1. Describe the gross anatomy of the kidney and its tissue layers.
2. Identify the parts of a nephron and the layers of the filtration membrane.
3. Trace the blood supply of the kidney and the path of urine.

WATCH AFTER YOU READ THIS CHAPTER

Concept videos: The urinary system

drsrennie-stack.github.io/new-build-bio4-solano/urinary-concept-videos.html

The Urinary System, an Overview

The urinary system filters the blood and makes urine. The kidneys do the work; the rest of the system carries and stores the urine.

THE PARTS OF THE SYSTEM AND WHAT EACH ONE DOES

Structure	What it is
Urinary system	The kidneys, ureters, urinary bladder, and urethra
Kidneys	The paired organs that filter the blood and form urine
Ureters	The paired tubes that carry urine from the kidneys to the bladder
Urinary bladder	The muscular sac that stores urine
Urethra	The tube that carries urine from the bladder to the outside
Roles of the kidney	Filtering wastes from the blood, balancing water, ions, and pH, and helping regulate blood pressure

The Kidneys, Gross Anatomy

The kidneys are bean-shaped organs against the back wall of the abdomen. Three tissue layers cushion and anchor each one, and the interior sorts into a cortex and a medulla.

THE POSITION, THE COVERINGS, AND THE REGIONS INSIDE

Structure	What it is
Location	Retroperitoneal, between T12 and L3; the right kidney sits slightly lower because of the liver
Renal hilum	The notch on the medial border where the ureter, vessels, and nerves enter and leave
Renal capsule	The smooth, fibrous inner layer that maintains the kidney's shape

Structure	What it is
Adipose capsule	The layer of fat around the capsule that cushions the kidney and holds it in place
Renal fascia	The outer layer of connective tissue that anchors the kidney to the abdominal wall
Renal cortex	The outer region of kidney tissue, just beneath the capsule
Renal columns	Extensions of the cortex that dip down between the renal pyramids
Renal medulla	The inner region of kidney tissue
Renal pyramids	The cone-shaped masses of the medulla, with their bases toward the cortex
Renal papilla	The tip of a renal pyramid, which drains urine into a minor calyx
Parenchyma	The functional tissue of the kidney, the cortex and pyramids together, which holds the nephrons

The Nephron

The nephron is the functional unit of the kidney, the structure that actually makes urine. It has two main parts: the renal corpuscle and the renal tubule.

THE PARTS OF A NEPHRON

Structure	What it is
Nephron	The microscopic functional unit of the kidney; each kidney holds about one million
Renal corpuscle	The filtering head of the nephron; the glomerulus inside the glomerular capsule
Glomerulus	The tuft of leaky, fenestrated capillaries where the blood is filtered
Glomerular capsule	Bowman's capsule; the cup that surrounds the glomerulus and collects the filtrate
Podocytes	The cells of the inner, visceral layer of the capsule, whose foot-like processes form filtration slits
Capsular space	The space inside the capsule where the filtrate collects before entering the tubule
Proximal convoluted tubule	The first, coiled segment of the renal tubule, lined with a brush border of microvilli
Nephron loop	The loop of Henle; the U-shaped segment with a descending limb and an ascending limb
Distal convoluted tubule	The second coiled segment of the renal tubule
Collecting duct	The duct that gathers fluid from many nephrons and carries it toward the renal papilla

KEY POINT

The segments of the renal tubule are named in the order the filtrate meets them: capsular space, proximal convoluted tubule, nephron loop, distal convoluted tubule, collecting duct. Learn the segments in that order and the path of the filtrate through the nephron comes free.

Nephrons come in two types, sorted by how deep their loop reaches.

COMPARE THEM

Type	Features
Cortical nephrons	About eighty five percent of nephrons; short loops that barely enter the medulla, wrapped by peritubular capillaries
Juxtamedullary nephrons	About fifteen percent of nephrons; long loops that reach deep into the medulla, served by the vasa recta; they concentrate the urine

The Juxtaglomerular Apparatus and Filtration Membrane

Where the distal tubule touches its own glomerulus, a small structure monitors and adjusts filtration. The filtration membrane itself is a three-layer sieve.

THE JUXTAGLOMERULAR APPARATUS

Structure	What it is
Juxtaglomerular apparatus	The structure where the distal convoluted tubule contacts the arterioles of its own renal corpuscle
Macula densa	The cells of the distal tubule that sense the sodium and chloride concentration of the fluid
Juxtaglomerular cells	Modified smooth muscle cells in the afferent arteriole that release renin when blood pressure is low

Blood is filtered across a three-layer membrane.

COMPARE WHAT EACH LAYER HOLDS BACK

Layer	What it holds back
Fenestrated endothelium	The pored wall of the glomerular capillary; it lets plasma pass but blocks blood cells
Basement membrane	The layer between the capillary and the podocytes; it blocks large plasma proteins
Slit membranes	The filtration slits between the podocyte processes; they block most of the smaller proteins that get past the basement membrane

The Blood Supply of the Kidney

Blood reaches the nephrons through a branching chain of arteries and leaves through a matching set of veins. Trace the arterial path.

1. **Renal artery** branches from the abdominal aorta and enters at the hilum
2. **Segmental and interlobar arteries** carry blood between the renal pyramids
3. **Arcuate arteries** arch along the bases of the pyramids, between the cortex and medulla
4. **Interlobular arteries** branch up into the cortex
5. **Afferent arteriole** delivers blood into the glomerulus
6. **Glomerulus** the capillary tuft where filtration occurs

7. **Efferent arteriole** carries blood out of the glomerulus
8. **Peritubular capillaries and vasa recta** wrap the renal tubules, then drain into veins that lead to the renal vein

KEY POINT

Blood meets two capillary beds in the kidney, one after the other: the glomerulus, where filtration happens, and then the peritubular capillaries and vasa recta around the tubules. The efferent arteriole is the vessel between them.

The Ureters, Bladder, and Urethra

Once urine leaves the nephrons it is collected, carried, stored, and finally released by the rest of the system.

FROM THE RENAL PAPILLA TO THE OUTSIDE

Structure	What it is
Minor and major calyces	The cup-shaped chambers that collect urine from the renal papillae and channel it into the renal pelvis
Renal pelvis	The funnel-shaped cavity that collects urine and passes it into the ureter
Ureters	The muscular tubes that carry urine from the kidneys to the bladder by peristalsis
Trigone	The smooth triangular area at the base of the bladder, marked by the two ureter openings and the urethral opening
Detrusor muscle	The smooth muscle layer of the bladder wall that contracts to empty it
Internal urethral sphincter	The involuntary smooth muscle ring at the bladder outlet
External urethral sphincter	The voluntary skeletal muscle ring that gives conscious control over urination
Male urethra	Longer; it passes through the prostate as the prostatic urethra, through the perineum as the membranous urethra, and through the penis as the spongy urethra
Female urethra	Shorter; it opens between the clitoris and the vaginal opening

The Path of Urine

Urine leaves the nephron and flows through a fixed series of structures to the outside.

1. **Collecting duct** gathers the fluid from many nephrons
2. **Minor and major calyces** collect the urine draining from the renal papillae
3. **Renal pelvis** funnels the urine toward the ureter
4. **Ureter** carries the urine down to the bladder
5. **Urinary bladder** stores the urine
6. **Urethra** carries the urine out of the body

Common Disorders of the Urinary System

Compare the common disorders of the urinary system by the structure each one affects.

COMMON DISORDERS OF THE URINARY SYSTEM

Disorder	What it is
Urinary tract infection	An infection of the urethra, bladder, or kidneys, usually by bacteria
Kidney stones	Hardened mineral deposits that form in the kidney and can lodge in the ureter
Glomerulonephritis	Inflammation of the glomeruli that impairs filtration
Renal failure	Loss of the kidney's ability to filter the blood, acute or chronic
Nephroptosis	A floating kidney; the kidney drops from its normal position when its fat support is lost
Urinary incontinence	Loss of voluntary control over urination, as stress, urge, overflow, or functional incontinence
Polycystic kidney disease	An inherited disorder in which many fluid-filled cysts enlarge the kidneys

POSITION AND PATHWAY PREDICT THE PROBLEM

The kidney depends on its **adipose capsule** to stay in place. When that fat is lost, the kidney can drop, kinking the ureter and blocking the flow of urine. The anatomy of the support explains the symptom.

Urine travels one narrow path. A **kidney stone** that forms in the renal pelvis causes no pain until it enters the **ureter**, where it scrapes and blocks a tube only millimeters wide. The location along that path predicts where the pain is felt.

The two urethras differ, and so does the risk. The **female urethra** is short, which gives bacteria a shorter climb to the bladder and is one reason urinary tract infections are more common in women.